Design and Calibration of a Digital Range Finder using ESP32

Atiqur Rahman (2202129), Sorder Rakib Hassan (2202140)
Department of Electrical and Electronic Engineering
Chittagong University of Engineering & Technology (CUET)

***Abstract*—A non-contact digital range finder was developed using an ESP32 microcontroller and an ultrasonic transducer. The instrument generates short bursts of ultrasound, measures the echo time of flight (ToF) and converts this to distance. A real-time distance (10–200 cm) is displayed on an I²C liquid crystal display (LCD). Because the speed of sound depends on temperature and, to a lesser extent, humidity, this dependence is discussed in the measurement model. Signal processing techniques such as moving-average filtering and Kalman filtering were explored to reduce measurement noise and improve accuracy. Calibration was carried out by comparing the sensor's output with a standard measuring tape at various distances, and performance metrics including linearity and error percentage were computed.**

I. Introduction

Accurate distance measurement is central to many electrical and electronic engineering applications, ranging from robotics and industrial automation to fluid-level control and non-contact gauging. Ultrasonic rangefinders provide a low-cost, robust method to make such measurements because they are insensitive to ambient light and electromagnetic interference [1]. These sensors operate by emitting a short burst of high-frequency sound and measuring the time elapsed until an echo returns from a target. Non-contact distance measurement avoids mechanical wear and is safer for applications involving moving parts or hazardous environments.

The ESP32 microcontroller family integrates a dual-core 32-bit processor, Wi-Fi, Bluetooth and a rich set of peripherals including UART, SPI, I²C and ADC interfaces. These modules operate from 3.0–3.6 V, have up to 26 GPIO pins and can clock at up to 240 MHz. Their high computing power and low power consumption make them suitable as the control unit for a digital range finder. An I²C LCD provides a simple user interface to display the measured distance [2].

This project aims to design and build a digital range finder that uses ultrasonic time-of-flight measurement with the ESP32 and to calibrate the device over a 10–100 cm range. The report describes the physics of ultrasonic ranging, factors affecting measurement accuracy and advanced signal-processing techniques, then outlines an action plan for implementing and calibrating the instrument.

II. Measurement Principle and Mode

Ultrasonic ranging can be implemented using several methods, including pulse-echo time-of-flight (ToF), phase-shift-based measurement, and frequency-modulated/chirp-based ranging approaches [1]. Each method has different trade-offs in hardware complexity, processing requirements, and noise robustness.

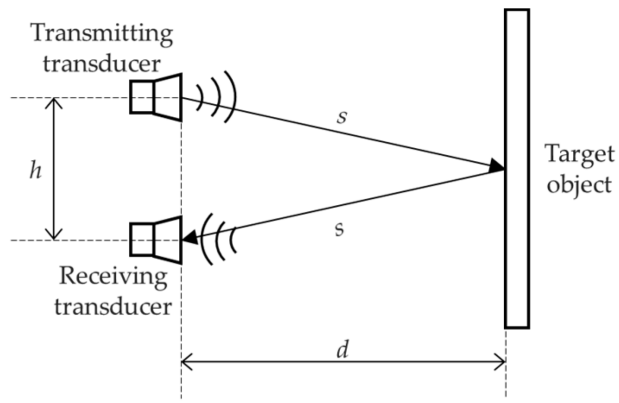

Fig. 1. Pitch-catch ultrasonic ranging geometry (adapted from [1]).

In this project, we employed the ToF method because of its simple implementation with the ESP32 platform. The system was configured in a pitch-catch mode, where separate transmitting and receiving transducers are used.

A. Time-of-Flight Principle

For a generic pulse-echo ToF configuration, distance is computed from travel time as

$$d = \frac{c\,t}{2} \tag{1}$$

where c is the speed of sound in air and t is the measured round-trip time.

For the pitch-catch geometry used in this work, Fig. 1 shows two separated transducers with spacing h. If the total ToF is t, then each acoustic path has length

$$s = \frac{c\,t}{2} \tag{2}$$

Using the right-triangle relation in the figure, the forward standoff distance to the target is

$$d = \sqrt{s^2 - \left(\frac{h}{2}\right)^2} = \sqrt{\left(\frac{c\,t}{2}\right)^2 - \left(\frac{h}{2}\right)^2} \tag{3}$$

When $h \ll 2d$, this reduces approximately to $d \approx \frac{c\,t}{2}$.

B. Speed of Sound in Air and Environmental Compensation

The speed of sound in air depends primarily on the ambient temperature. In dry air at $0°$C, the speed of sound is approximately

$$c \approx 331.45\ \mathrm{m\,s^{-1}} \tag{4}$$

A commonly used linear approximation relating sound speed to air temperature is [1]

$$c \approx 331.45 + 0.607\,T_C \tag{5}$$

where T_C is the air temperature in degrees Celsius. This relation indicates that the speed of sound increases by about $0.6\,\mathrm{m\,s^{-1}}$ for every $1°$C increase in temperature.

Environmental factors such as humidity and atmospheric pressure also affect the propagation speed of sound, though their influence is comparatively smaller. Across typical ranges of 0–100% relative humidity and 50–110 kPa pressure, the variation in sound speed is generally less than 0.7% [1].

For improved measurement accuracy, the ESP32 can update distance calculations using a temperature-dependent value of c. If ambient temperature is known, systematic errors caused by environmental variations can be reduced.

III. Methodology and Design

A. Hardware Implementation

The prototype consists of the following main components:

- ESP32 development board
- HC-SR04 ultrasonic sensor
- Low-power laser module mounted at the centerline of the ultrasonic transducers
- I^2C 16×2 LCD module with backpack
- Breadboard, jumper wires, and regulated power supply

1) Circuit Connections: The ESP32 development board is powered using a regulated 3.3 V supply. The ultrasonic sensor module is powered from the ESP32 supply rail, and its ground is connected to the ESP32 ground to establish a common reference.

The Trigger pin of the ultrasonic sensor is connected to a digital GPIO pin configured as an output. The Echo pin is connected to a GPIO pin configured as an input.

The I^2C LCD module is connected to the ESP32 I^2C interface. The SDA and SCL lines are connected to GPIO 21 and GPIO 22 respectively.

The laser module is mechanically fixed so that its beam emerges from the middle of the two ultrasonic transducers. This laser is used only as a visual pointer to align the measurement axis with the target point.

2) Prototype Assembly: All components are assembled on a breadboard during the prototyping stage. The ultrasonic transducer is positioned so that it has a clear line of sight to the target surface. The laser pointer is adjusted during assembly so its spot coincides with the center of the ultrasonic sensing direction. Care is taken to minimize nearby reflective objects that could introduce unwanted echoes. If necessary, simple shielding or damping materials can be placed around the sensor to reduce acoustic reflections from surrounding surfaces.

B. Software Flow

The ESP32 generates a trigger pulse sequence for each measurement cycle. First, the trigger pin is set LOW for $2\,\mu$s, then set HIGH for $10\,\mu$s, and then returned LOW. The echo pulse width is measured using the Arduino `pulseIn()` function on the echo pin with a timeout of $30{,}000\,\mu$s.

If no valid echo pulse is received within the timeout, that cycle is ignored. If a valid pulse is received, the time value is stored and the distance is computed using Eq. (3). The resulting distance is then sent to the active user interface mode.

Two software operating modes are implemented:

- **LCD mode (local mode):** after powering the ESP32, pressing the BOOT button once starts local display mode, where distance is shown on the LCD.
- **Website mode (Wi-Fi mode):** after reset, pressing the BOOT button twice starts web mode. The ESP32 starts the web interface and displays its IP address. A user connects over Wi-Fi, opens the IP address in a browser, and can view live readings, record readings, and perform calibration tasks.

IV. Calibration and Results

A. Calibration Procedure

Explain how you compared your device with the standard meter. Describe the test environment and setup conditions (e.g., fixed distances, target material, and measurement alignment).

B. Data and Analysis

[Populate the table below with your experimental data.]

TABLE I
Comparison: Prototype vs. Standard Meter

Exp. No.	Standard Reading	ESP32 Reading	% Error
1	[Value]	[Value]	[Calc %]
2	[Value]	[Value]	[Calc %]
3	[Value]	[Value]	[Calc %]
4	[Value]	[Value]	[Calc %]
5	[Value]	[Value]	[Calc %]

C. Performance Graph

[Insert a plot showing Standard vs. Measured values. Discuss the linearity of the slope.]

V. Error Analysis and Discussion

Discuss the sources of error. Was it due to ADC resolution? Environmental noise? Sensor lag? Explain how the calibration factor improved your readings.

VI. Conclusion

Summarize the success of the prototype and suggest improvement for future iterations.

REFERENCES

[1] Z. Qiu, Y. Lu, and Z. Qiu, "Review of ultrasonic ranging methods and their current challenges," *Micromachines*, vol. 13, no. 4, 2022. [Online]. Available: https://www.mdpi.com/2072-666X/13/4/520

[2] Espressif Systems, *ESP32 Series Datasheet*, 2023. [Online]. Available: https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf